

TU	SOL	LUA	Lat.	CREP	SOL	Náut.	Civil	Náut.	LUA - Pôr	20	21	22	23
h m	h m	h m	h m	h m	h m	h m	h m	h m	h m	h m	h m	h m	h m
20 00	178 24.2 N20 36.6	187 55.2 6.6 N23 53.3	1.9 57.7	N 72	N 70	N 72	N 70	N 72	N 70	N 72	N 70	N 72	N 70
01	193 24.2 36.1	202 20.8 6.6	23 51.4 2.0	57.7	N 70	N 72	N 70	N 72	N 70	N 72	N 70	N 72	N 70
02	208 24.2 35.6	216 46.4 6.5	23 49.4 2.1	57.7	N 70	N 72	N 70	N 72	N 70	N 72	N 70	N 72	N 70
03	223 24.1 35.2	231 11.9 6.5	23 47.3 2.3	57.8	N 70	N 72	N 70	N 72	N 70	N 72	N 70	N 72	N 70
04	238 24.1 34.7	245 37.4 6.6	23 45.0 2.5	57.8	N 70	N 72	N 70	N 72	N 70	N 72	N 70	N 72	N 70
05	253 24.1 34.2	260 03.0 6.5	23 42.5 2.6	57.8	N 70	N 72	N 70	N 72	N 70	N 72	N 70	N 72	N 70
06	268 24.0 N20 33.7	274 28.5 6.5	N23 39.9 2.7	57.8	N 58	N 70	N 58	N 70	N 58	N 70	N 58	N 70	N 58
07	288 24.0 33.3	288 54.0 6.5	23 37.2 2.9	57.9	N 58	N 70	N 58	N 70	N 58	N 70	N 58	N 70	N 58
08	298 24.0 32.8	303 19.5 6.4	23 34.3 3.1	57.9	N 58	N 70	N 58	N 70	N 58	N 70	N 58	N 70	N 58
09	313 23.9 32.3	317 44.9 6.5	23 31.2 3.2	57.9	N 58	N 70	N 58	N 70	N 58	N 70	N 58	N 70	N 58
10	328 23.9 31.8	332 10.4 6.5	23 28.0 3.3	57.9	N 58	N 70	N 58	N 70	N 58	N 70	N 58	N 70	N 58
11	343 23.9 31.3	346 35.9 6.5	23 24.7 3.5	58.0	N 58	N 70	N 58	N 70	N 58	N 70	N 58	N 70	N 58
12	358 23.8 N20 30.9	1 01.4 6.4	N23 21.2 3.7	58.0	N 40	N 70	N 40	N 70	N 40	N 70	N 40	N 70	N 40
13	13 23.8 30.4	15 26.8 6.5	23 17.5 3.7	58.0	N 35	N 70	N 35	N 70	N 35	N 70	N 35	N 70	N 35
14	28 23.8 29.9	29 52.3 6.5	23 13.8 4.0	58.0	N 30	N 70	N 30	N 70	N 30	N 70	N 30	N 70	N 30
15	43 23.7 29.4	44 17.8 6.5	23 09.8 4.1	58.1	N 20	N 70	N 20	N 70	N 20	N 70	N 20	N 70	N 20
16	58 23.7 28.9	58 43.3 6.5	23 05.7 4.2	58.1	N 10	N 70	N 10	N 70	N 10	N 70	N 10	N 70	N 10
17	73 23.7 28.5	73 08.8 6.5	23 01.5 4.4	58.1	N 0	N 70	N 0	N 70	N 0	N 70	N 0	N 70	N 0
18	88 23.6 N20 28.0	87 34.3 6.5	N22 57.1 4.5	58.1	N 10	N 70	N 10	N 70	N 10	N 70	N 10	N 70	N 10
19	103 23.6 27.5	101 59.8 6.5	22 52.6 4.7	58.2	N 20	N 70	N 20	N 70	N 20	N 70	N 20	N 70	N 20
20	118 23.6 27.0	116 25.3 6.5	22 47.9 4.8	58.2	N 30	N 70	N 30	N 70	N 30	N 70	N 30	N 70	N 30
21	133 23.5 26.5	130 50.8 6.5	22 43.1 5.0	58.2	N 35	N 70	N 35	N 70	N 35	N 70	N 35	N 70	N 35
22	148 23.5 26.0	145 16.3 6.6	22 38.1 5.1	58.2	N 40	N 70	N 40	N 70	N 40	N 70	N 40	N 70	N 40
23	163 23.5 25.6	159 41.9 6.6	22 33.0 5.2	58.3	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
00	178 23.5 N20 25.1	174 07.5 6.6	N22 27.8 5.4	58.3	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
01	193 23.4 24.6	188 33.1 6.6	22 22.4 5.6	58.3	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
02	208 23.4 24.1	202 58.7 6.6	22 16.8 5.7	58.3	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
03	223 23.4 23.6	217 24.3 6.6	22 11.1 5.8	58.4	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
04	238 23.3 23.1	231 49.9 6.7	22 05.3 6.0	58.4	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
05	253 23.3 22.6	246 15.6 6.7	21 59.3 6.1	58.4	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
06	268 23.3 N20 22.1	260 41.3 6.7	N21 53.2 6.2	58.4	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
07	283 23.3 21.7	275 07.0 6.8	21 47.0 6.4	58.5	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
08	298 23.2 21.2	289 32.8 6.7	21 40.6 6.6	58.5	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
09	313 23.2 20.7	303 58.5 6.8	21 34.0 6.6	58.5	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
10	328 23.2 20.2	318 24.3 6.9	21 27.4 6.8	58.5	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
11	343 23.1 19.7	332 50.2 6.8	21 20.6 7.0	58.5	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
12	358 23.1 N20 19.2	347 16.0 6.9	N21 13.6 7.1	58.6	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
13	18 23.1 18.7	1 41.9 7.0	21 06.5 7.2	58.6	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
14	28 23.1 18.2	16 07.9 6.9	20 59.3 7.3	58.6	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
15	43 23.0 17.7	30 33.8 7.0	20 52.0 7.5	58.6	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
16	58 23.0 17.2	44 59.8 7.0	20 44.5 7.7	58.7	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
17	73 23.0 16.7	59 25.8 7.1	20 36.8 7.7	58.7	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
18	88 23.0 N20 16.2	73 51.9 7.1	N20 29.1 7.9	58.7	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
19	103 22.9 15.7	88 18.0 7.2	20 21.2 8.0	58.7	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
20	118 22.9 15.2	102 44.2 7.1	20 13.2 8.2	58.7	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
21	133 22.9 14.7	117 10.3 7.3	20 05.0 8.2	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
22	148 22.9 14.2	131 36.6 7.2	19 56.8 8.4	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
23	163 22.8 13.7	146 02.8 7.3	19 48.4 8.6	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
00	178 22.8 N20 13.2	160 29.1 7.4	N19 39.8 8.6	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
01	193 22.8 12.7	174 55.5 7.4	19 31.2 8.8	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
02	208 22.8 12.2	189 21.9 7.4	19 22.4 8.9	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
03	223 22.7 11.7	203 48.3 7.5	19 13.5 9.0	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
04	238 22.7 11.2	218 14.8 7.6	19 04.5 9.2	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
05	253 22.7 10.7	232 41.4 7.5	18 55.3 9.3	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
06	268 22.7 N20 10.2	247 07.9 7.6	N18 46.0 9.3	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
07	283 22.7 09.7	261 34.5 7.7	18 36.7 9.6	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
08	298 22.6 09.2	276 01.2 7.7	18 27.1 9.6	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
09	313 22.6 08.7	290 27.9 7.8	18 17.5 9.7	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
10	328 22.6 08.2	304 54.7 7.8	18 07.8 9.9	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
11	343 22.6 07.7	319 21.5 7.9	17 57.9 9.9	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
12	358 22.5 N20 07.2	333 48.4 7.9	N17 48.0 10.1	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
13	13 22.5 06.7	348 15.3 7.9	17 37.9 10.2	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
14	28 22.5 06.2	2 42.2 8.1	17 27.7 10.3	59.1	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
15	43 22.5 05.7	17 09.3 8.0	17 17.4 10.4	59.1	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
16	58 22.5 05.1	31 36.3 8.1	17 07.0 10.5	59.1	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
17	73 22.4 04.6	46 03.4 8.2	16 56.5 10.7	59.1	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
18	88 22.4 N20 04.1	60 30.6 8.2	N16 45.8 10.7	59.1	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
19	103 22.4 03.6	74 57.8 8.3	16 35.1 10.8	59.1	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
20	118 22.4 03.1	89 25.1 8.3	16 24.3 11.0	59.1	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
21	133 22.4 02.6	103 52.4 8.3	16 13.3 11.0	59.2	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
22	148 22.4 02.1	118 19.7 8.5	16 02.3 11.1	59.2	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
23	163 22.3 01.6	132 47.2 8.4	N15 51.2 11.3	59.2	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
00	178 22.3 N20 01.1	160 29.1 7.4	N19 39.8 8.6	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
01	193 22.3 00.6	174 55.5 7.4	19 31.2 8.8	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
02	208 22.3 00.1	189 21.9 7.4	19 22.4 8.9	58.8	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
03	223 22.3 00.0	203 48.3 7.5	19 13.5 9.0	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
04	238 22.3 00.0	218 14.8 7.6	19 04.5 9.2	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
05	253 22.3 00.0	232 41.4 7.5	18 55.3 9.3	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
06	268 22.3 00.0	247 07.9 7.6	N18 46.0 9.3	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
07	283 22.3 00.0	261 34.5 7.7	18 36.7 9.6	58.9	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
08	298 22.3 00.0	276 01.2 7.7	18 27.1 9.6	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
09	313 22.3 00.0	290 27.9 7.8	18 17.5 9.7	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
10	328 22.3 00.0	304 54.7 7.8	18 07.8 9.9	59.0	N 50	N 70	N 50	N 70	N 50	N 70	N 50	N 70	N 50
11	343 22.3 00.0	319 21.5 7.											